EE561 Advanced Static Power Conversion · Multilevel Converters module

Potential Final Questions: Building the Levels

Proposed by Sefer İdacı for the final examination

Two questions for a closed-book exam with the instructor's cheat sheet, each designed for about 20 minutes and 25 points, on the topologies talk: device stress in an NPC leg, and the model and design of a flying-capacitor leg. Every question asks for a derivation, a numerical result and an engineering interpretation; full solutions and a marking guide follow on the later pages. All numbers differ from the lecture examples.

Question 1 · Device stress in a three-level NPC leg

ANALYSIS — **25 pts**

A 3-level NPC leg with V_{dc} = 800 V uses sine-triangle PWM at unity power factor: $r = m \sin\theta$, $i_a = \hat{I} \sin\theta$, with m = 0.8, $\hat{I}$ = 80 A, f_{sw} = 8 kHz. Device models: switch $v = 0.9\ \text{V} + 10\ \text{m}\Omega \cdot i$; diode $v = 1.0\ \text{V} + 8\ \text{m}\Omega \cdot i$; turn-on plus turn-off energy of a switch $E_{on} + E_{off} = 2.5 \times 10^{-7}\ \text{J/(V·A)} \times V \times I$; diode recovery energy $E_{rr} = 0.6 \times 10^{-7}\ \text{J/(V·A)} \times V \times I$.

a. For the positive half-cycle, state which devices conduct in states P and O, and identify the commutating pair. [5]

b. Derive the average and RMS currents of S1, S2 and the clamp diode D5 as functions of m and $\hat{I}$, and evaluate them. [10]

c. Compute the conduction and switching losses of S1, S2 and D5. Which device is the hottest? [6]

d. Repeat the ranking for m = 0.2 (a motor at low speed) and explain the change. How can an active NPC (ANPC) leg relieve the inner switches when the same converter works as a rectifier? [4]

Question 2 · Flying-capacitor leg: model and design

ANALYSIS AND DESIGN — **25 pts**

The 3-level flying-capacitor leg in the figure has V_{dc} = 1000 V and a flying-capacitor target V_{fc}^* = 500 V. Switches S1/S1′ and S2/S2′ are complementary pairs.

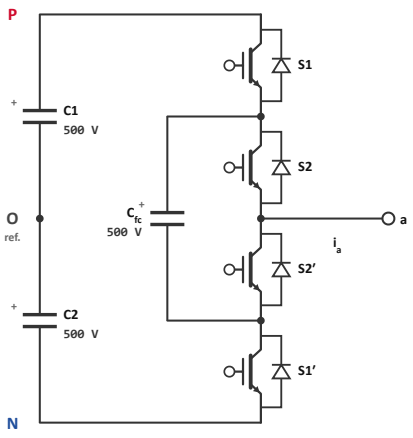
a. Tabulate the four states (S1 S2 = 11, 10, 01, 00) with the pole voltage v_{aO} in terms of V_{dc} and V_{fc}, and the capacitor current i_{fc} in terms of i_a. [6]

b. Write v_{aO} and i_{fc} as switching functions, average them over one switching period with duties d_1, d_2, and show that phase-shifted carriers ($d_1 = d_2$) give zero average capacitor current. [6]

c. With phase-shifted carriers a redundant state lasts at most half a switching period. Size C_{fc} for $\Delta V_{fc,pp} \le 10$ V at $\hat{I}$ = 60 A and f_{sw} = 20 kHz. [6]

d. V_{fc} is measured as 470 V while $i_a = -30$ A and the next output level is zero. Which state do you apply, and what pole voltage does it produce? [4]

e. Why must C_{fc} be pre-charged before PWM starts? [3]



3-level flying-capacitor leg (O: reference midpoint only).

Solutions and marking guide

Solution 1 · Device stress in a three-level NPC leg — 25 pts

(a) Positive half-cycle ($i_a > 0$): P → S1 + S2; O → D5 + S2. The commutating pair is **S1 ↔ D5** (short loop C1–S1–D5); S2 stays on and conducts the whole half-cycle. [5]

(b) With duties $d_P = m \sin\theta$ and $d_O = 1 - m \sin\theta$ over $0 < \theta < \pi$:

$$I_{avg,S1} = (1/2\pi)\int_0^{\pi} m \sin\theta \cdot \hat{I} \sin\theta \, d\theta = m\hat{I}/4, \quad {I_{rms,S1}}^2 = (1/2\pi)\int_0^{\pi} m \sin\theta \cdot \hat{I}^2 \sin^2\theta \, d\theta = 2m\hat{I}^2/3\pi$$

$$I_{avg,S2} = \hat{I}/\pi, \quad {I_{rms,S2}}^2 = \hat{I}^2/4; \quad I_{avg,D5} = \hat{I}(1/\pi - m/4), \quad {I_{rms,D5}}^2 = \hat{I}^2(1/4 - 2m/3\pi)$$

	I_{avg}	I_{rms}	conduction	switching	total
S1	16.0 A	33.0 A	25.3 W	20.4 W	**45.6 W**
S2	25.5 A	40.0 A	38.9 W	0	38.9 W
D5	9.46 A	22.7 A	13.6 W	4.9 W	18.5 W

(c) $P_{cond} = V_0 I_{avg} + r\,{I_{rms}}^2$. Switching occurs in every carrier period of the positive half-cycle at the commutation voltage $V_{dc}/2 = 400$ V: $P_{sw,S1} = f_{sw} \cdot 2.5\times10^{-7} \cdot 400 \cdot \hat{I}/\pi = 20.4$ W, $P_{rr,D5} = f_{sw} \cdot 0.6\times10^{-7} \cdot 400 \cdot \hat{I}/\pi = 4.9$ W. Hottest: **S1 (45.6 W)**. [6]

(d) At $m = 0.2$: S1 = 6.3 + 20.4 = 26.7 W, D5 = 32.1 + 4.9 = 37.0 W, S2 = 38.9 W (unchanged). The zero-state duty $1 - m \sin\theta$ grows, so the inner switch and the clamp diode now carry most of the current; D5 overtakes S1. In rectifier operation the commutation moves to S3 ↔ D1, so the inner switches both conduct and switch; in an ANPC leg the zero state can use the active clamps (O_U/O_L paths) so that S5/S6 take the commutation and the loss is redistributed. [4]

Marking (b): S1 derivation 4, S2 3, D5 3 (2 for the formula, 1 for the number).

Solution 2 · Flying-capacitor leg — 25 pts

(a) [6]

S1 S2	v_{aO}	nominal	i_{fc} (into + terminal)
11	$+V_{dc}/2$	+500 V	0
10	$V_{dc}/2 - V_{fc}$	0 V	$+i_a$
01	$-V_{dc}/2 + V_{fc}$	0 V	$-i_a$
00	$-V_{dc}/2$	−500 V	0

(b) $v_{aO} = -V_{dc}/2 + S_1(V_{dc} - V_{fc}) + S_2 V_{fc}$, $i_{fc} = (S_1 - S_2)\, i_a$. Averaging: $\langle v_{aO}\rangle = -V_{dc}/2 + d_1(V_{dc} - V_{fc}) + d_2 V_{fc}$, $\langle i_{fc}\rangle = (d_1 - d_2)\, i_a$. Phase-shifted carriers with one reference give $d_1 = d_2 \rightarrow \langle i_{fc}\rangle = 0$ and $\langle v_{aO}\rangle = (2d - 1)V_{dc}/2$. [6]

(c) $\Delta V_{fc,pp} \le \hat{I}(T_s/2)/C_{fc} \rightarrow C_{fc} \ge \hat{I}/(2 f_{sw} \Delta V) = 60/(2 \cdot 20\,000 \cdot 10) =$ **150 µF** (voltage rating ≥ 500 V plus margin). [6]

(d) V_{fc} is low → we need $i_{fc} = (S_1 - S_2) i_a > 0$. With $i_a < 0$ this requires $S_1 - S_2 = -1$: **state 01**, $i_{fc} = +30$ A. It produces $v_{aO} = -500 + 470 =$ **−30 V** instead of 0. [4]

(e) With $V_{fc} = 0$, state 11 places $V_{dc} - V_{fc} = 1000$ V across S1′, a device rated for about $V_{dc}/2$: it would fail. A pre-charge circuit or a controlled start-up sequence brings V_{fc} to 500 V first. [3]